

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1703

Lindendale D.P 138 kV
October 15, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1703. The Interconnection Customer proposes a 147.4 MW solar plus storage facility interconnecting at the Lindendale D.P 138 kV substation in the Virginia area. The request was received on March 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 481 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,825,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
OCCOQUAN – CARTWRIGHT 138.0 kV Circuit 2	OCCOQUAN – CARTWRIGHT 138.0 kV L	104.9	260	15.8
CARTWRIGHT – OCCOQUAN 138.0 kV	CARTWRIGHT – OCCOQUAN SUBSTATION 138.0 kV Circuit 2	102.8	265	15.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
OCCOQUAN – CARTWRIGHT 138.0 kV #2	Reconductor line	\$1,494,000	73% (\$1,090,000)
CARTWRIGHT – OCCOQUAN 138.0 kV	Reconductor line	\$1,008,000	73% (\$735,000)
	Total allocated to project		\$1,825,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.